Enhancing Land Cover Mapping in Mixed Vegetation Regions Using Remote Sensing Evapotranspiration

Jie Wang, Zhenxin Bao, Amgad Elmahdi, Jianyun Zhang, Guoqing Wang, Cuishan Liu, and Houfa Wu

***Abstract*— Vegetation constitutes a significant portion of land cover. Due to the high spatial heterogeneity of site conditions and the similarities in spectral reflectance and shapes among different vegetation types, land cover mapping accuracy is often low in mixed regions with multiple vegetation types. In addition to traditional factors such as spectral characteristics, topography, and commonly used features, we present a novel land cover mapping framework that incorporates evapotranspiration, which exhibits significant variations among vegetation types. The proposed land cover mapping framework consists of the following steps: 1) estimating year-round actual evapotranspiration using the remote sensing SEABL model; 2) training a classifier to classify land cover based on integrated factors, including spectral bands, spectral indices, topography, night light data, and evapotranspiration; 3) generating land use and land cover mapping using evapotranspiration (ETLULC); and 4) comparing and evaluating the accuracy and variability of results against different input schemes and existing products. In a typical mixed region with multiple vegetation types, the ETLULC framework demonstrates impressive performance, achieving an overall accuracy of 93%. The classification accuracy for all land cover types exceeds 90%. Compared to traditional methodologies that do not incorporate evapotranspiration as an input, ETLULC significantly improves the recognition accuracy for cropland, forest, and grassland by 5.4%–15.3%, 0%–15.7%, and 3.0%–20.4%, respectively. Moreover, ETLULC exhibits strong agreement with existing products applied in the Ordos Basin, particularly for cropland (54.7%–82.3%), forest (32.2%–71.7%), and grassland (56.4%–94.3%). The performance of ETLULC underscores the effectiveness of this innovative land cover mapping framework. This study introduces a novel approach by leveraging the spatial heterogeneity of vegetation characterized by evapotranspiration to enhance the accuracy of land cover mapping. This method holds significant practical value and has broad applicability in identifying effective feature combinations for vegetation recognition in extensively distributed mixed vegetation regions.**

***Index Terms*— Evapotranspiration, land cover classification, remote sensing, vegetation.**

Manuscript received 7 October 2023; revised 28 February 2024; accepted 26 March 2024. Date of publication 1 April 2024; date of current version 7 May 2024. This work was supported in part by the National Key Research and Development Program of China under Grant 2022YFC3205200 and in part by the National Natural Science Foundation of China under Grant 41961124007 and Grant 52121006. *(Corresponding author: Zhenxin Bao.)*

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Digital Object Identifier 10.1109/TGRS.2024.3383217

I. Introduction

ACCURATE and large-scale land cover maps play a pivotal role in providing fundamental data support for environmental monitoring, policy-making, and ecological protection [1], [2]. Vegetation constitutes a significant portion of global land cover, encompassing more than two-thirds of the Earth's surface [3]. Accurate identification of vegetation types is crucial for land use classification, as vegetation plays a vital role in climate change, hydrological cycles, and ecological balance by connecting the atmosphere, hydrosphere, and pedosphere [4], [5]. Nature exhibits a diverse array of vegetation types, cross-distributed with strong spatial heterogeneity properties [6].

Various land use and land cover (LULC) data products have been released with diverse spatial and temporal resolutions, employing different inversion methods across distinct time periods [7], [8]. However, there exists significant inconsistency among LULC products due to differences in satellite operation mode, sensor type, classification method, and verification method [9]. Many experts have studied the consistency and postprocessing of spatial heterogeneity in various products [10], [11]. Traditional remote sensing interpretation algorithms often yield relatively low accuracy in classifying vegetation in mixed regions with multiple vegetation types due to their similar spectral reflectance and complex spatial distributions [12]. While ground surveys can effectively capture spatial patterns in highly heterogeneity areas, they are limited to selecting typical transects within the study area, hindering long-term monitoring and response to dynamic changes in driving factors [13]. Vegetation types, such as cropland, forest, and grassland, although distinguishable from other land cover types based on their distinct spectral characteristics, often exhibit subtle spectral differences among themselves, posing challenges for remote sensing-based classification [14], [15], [16]. The classification of land cover types in mixed vegetation areas remains a significant challenge, and improving classification accuracy in such regions is a current research difficulty.

In recent years, substantial progress has been made in vegetation classification using remote sensing data. However, it has not fully met the practical application requirements

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for vegetation classification [17]. Previous studies have proposed various methods to improve the accuracy of vegetation classification, broadly categorized into five types: 1) based on vegetation index; 2) using mixed pixel decomposition algorithm; 3) relying on multitemporal information; 4) integrating multisource LULC data; and 5) incorporating additional terrain, spectrum, and other supplementary information [3]. The classification based on vegetation index utilizes the high reflectivity of vegetation in the red and near-infrared (NIR) bands and establishes a vegetation index through mathematical operations to distinguish vegetation from nonvegetation [18], [19]. The mixed pixel decomposition algorithm is primarily applied to classify medium- to low-resolution satellite images, employing linear models and nonlinear models [20], [21]. Vegetation classification based on multitemporal remote sensing images takes into account the phenological characteristics of vegetation growth and development, considering spectral characteristics that evolve with changing seasons [22], [23]. Integrating multisource remote sensing data involves mathematically combining existing products to enhance classification accuracy, employing methods such as Bayesian averaging [24], decision tree [25], and Dempster–Shafer [14]. Furthermore, vegetation distribution is influenced by terrain, climate, soil, and other factors; thus, incorporating supplementary information can significantly enhance classification accuracy in vegetation remote sensing [26].

However, each method has its limitations. The vegetation index-based approach excels in distinguishing vegetation from nonvegetation [27]. The mixed pixel decomposition algorithm is more complex, limiting its efficiency in the exploratory stage of classification [28]. Methods relying on multitemporal information are constrained by the availability of long-term data acquisition [29]. The fusion of multisource LULC data introduces high uncertainty due to product and method selection. The method of adding supplementary information is relatively efficient, but the correlation between feature extraction and vegetation is pivotal for improving classification accuracy [30]. Previous studies have predominantly focused on examining the influence of environmental factors on vegetation, giving comparatively less attention to the reverse perspective—how vegetation distribution affects environmental factors. This consideration is crucial when selecting additional supplementary information for analysis [31]. Numerous environmental factors influence vegetation [32], [33]. However, it is essential to acknowledge that vegetation distribution can vary even within similar environmental conditions. Conversely, vegetation may also be similar in different environmental settings, limiting the efficacy of this method in achieving significant improvements.

Green vegetation is the primary determinant of land surface evapotranspiration [34]. It increases land surface roughness, accelerates turbulent water vapor exchange, and facilitates plant roots in absorbing soil moisture [35], [36]. Most studies emphasize that vegetation change is a key factor affecting evapotranspiration [37]. Evapotranspiration values vary dramatically across different vegetation types [38]. Under similar microclimatic conditions, vegetation with a larger leaf area typically exhibits a smaller total stomatal resistance, facilitating easier escape of water vapor from stomata into the canopy air [39]. In general, forests typically exhibit the highest evapotranspiration, followed by cropland and grassland [40]. The evapotranspiration of cropland and grassland is relatively similar. Some studies argue that cropland evapotranspiration is greater than that of grassland [39], while others suggest the opposite [41]. In addition, evapotranspiration across different vegetation types significantly varies in spring and summer [42], [43]. These studies demonstrate that the spatial variability of evapotranspiration can reflect differences in the distribution of vegetation types to some extent. However, there is a paucity of research on using evapotranspiration information to distinguish land cover vegetation types.

The Ordos Basin, situated in northern China and surrounded by mountainous terrain, exhibits a mixed spatial pattern of cropland–forest–grassland, typifying a region with mixed vegetation [44], [45]. The accuracy of existing publicly released products in this region has significantly decreased, failing to meet the requirements of basic research and practical applications [17], [46]. This article proposes an innovative method that it is possible to enhance classification accuracy in land cover mapping by considering the spatial heterogeneity of vegetation feedback to atmospheric water vapor, using actual evapotranspiration as supplementary information. The remainder of this article is structured as follows. In Section II, we introduce the study area and dataset. In Section III, we outline the land cover mapping framework and other methodologies, including the Surface Energy Balance Algorithm for Land (SEBAL) remote sensing model, classification system, input schemes, and accuracy assessment. Section IV presents results related to evapotranspiration calculation and validation, land cover classification with evapotranspiration, intercomparison with existing thematic products, and the relationship between land cover type and evapotranspiration. Section V discusses the advancements and limitations of the proposed method presented in this article and compares it with previous results. Finally, we conclude with a summary of the main findings and propose future research directions.

II. Study Area and Dataset

A. Study Area

The Ordos Basin (105°30′E–112°10′E, 34°30′N–40°30′N) is situated in the upper and middle reaches of the Yellow River Basin in China, spanning five provinces and regions including Inner Mongolia, Ningxia, Gansu, Shaanxi, and Shanxi, encompassing a total area of approximately 380 000 km^2 (see Fig. 1). The basin is encircled by mountain ranges, namely, the Qinling Mountains to the south, the Yinshan Mountains to the north, the Lüliang Mountains to the east, and the Helan Mountains–Liupan Mountains to the west. Within the basin, the Hetao Plain, Yinchuan Basin, and Weinan Basin are distributed from north to south. The topography generally declines from northwest to southeast, displaying significant fluctuations in certain areas. The Ordos Basin lies within the typical arid-semiarid and continental monsoon climate

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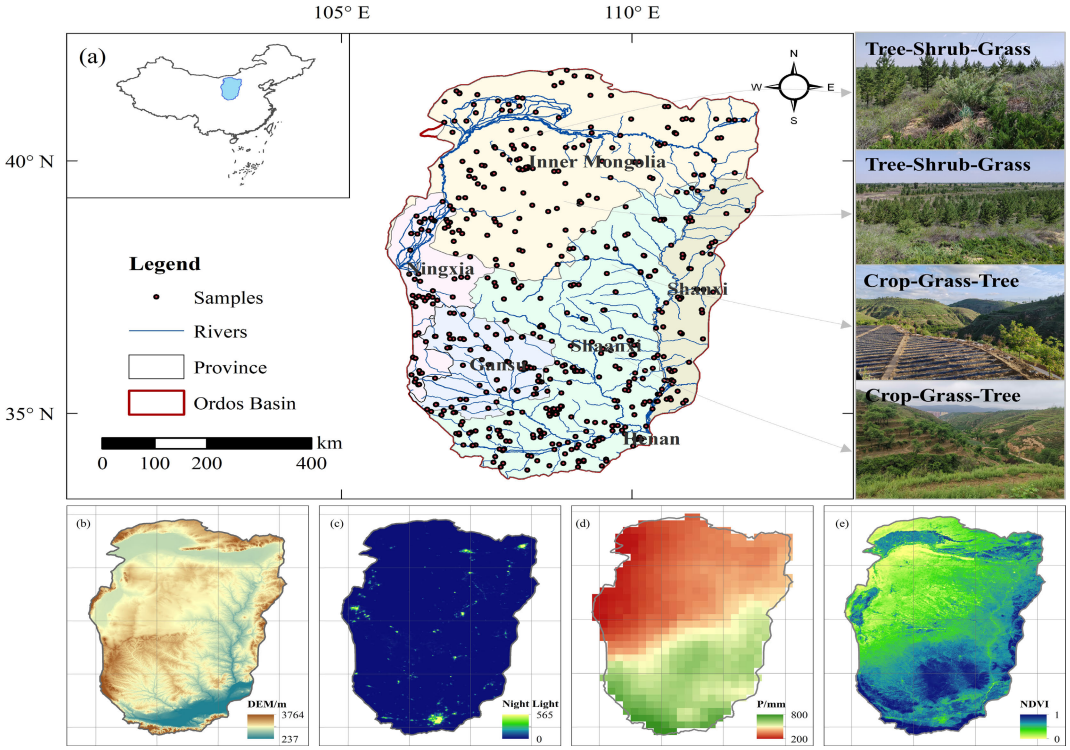


Fig. 1. Locations and characteristics of the studied area and the typical field pictures of mixing vegetation in Ordos Basin. (a) Location of the study area. (b)–(e) DEM, night light, annual mean precipitation, and NDVI, respectively.

zone, characterized by an average annual precipitation ranging from 100 to 800 mm and an annual mean temperature between 2 °C and 10 °C [47]. The climate is primarily influenced by latitude, monsoon circulation, distance from the ocean, and terrain variations, leading to climate disparities across different regions within the study area due to its extensive coverage and wide geographical span.

The vegetation distribution in the Ordos Basin follows a zonal pattern from southeast to northwest, originating from complex topography and climate [15]. The northern is predominantly covered by sand, hosting various vegetation types, including *Sabina vulgaris*, *Pinus sylvestris*, *Artemisia sphaerocephala*, *Salix matsudana*, *Caragana korshinskii*, and so on. The southern part corresponds to the Loess Plateau and is characterized by the prevailing vegetation type of *Platycladus orientalis*, poplar, and so on. Moreover, this region holds agricultural significance, with major crop types such as millet, wheat, and corn. The vegetation landscape pattern exhibits pronounced spatial heterogeneity, with a mix of multiple vegetation types influenced by topographical features [6]. The vegetation field survey image in Fig. 1 illustrates this heterogeneity, showcasing a tree–shrub–grass mix in the northern regions and a crop–grass–tree mix in the southern parts. Specifically, the first two pictures indicate a mixing of *Sabina vulgaris*, *Hedysarum mongolicum*, *Artemisia ordosica*, *Pinus sylvestris*, and other herbaceous plants. The last two pictures demonstrate vegetation mixing by cereals, fruit trees, oriental arborvitae, and herbaceous. The distinct vegetation types are a result of varying ecological restoration and control policies and projects, such as grain for green and grassland [44], [48]. These initiatives have significantly enhanced vegetation coverage and effectively controlled soil erosion [49]. However, due to the rugged topography and high spatial heterogeneity in vegetation distribution, in conjunction with extensive ecological engineering efforts, existing remote sensing inversion products struggle to accurately represent the vegetation's spatial patterns within the mixed landscapes of trees, croplands, and grass/shrubs. Consequently, the Ordos Basin serves as a representative area for vegetation classification.

B. Dataset

Multispectral imaging satellites have gained widespread recognition as an ideal data source for land cover classification [27], [50]. Notable examples include the Landsat and Sentinel satellites. Landsat 8, an updated iteration of the Landsat series, is equipped with the Operational Land Imager (OLI) for collecting surface reflectance (SR) data [51]. From the Google Earth Engine (GEE) Platform, we acquired six spectral bands: blue (B2), green (B3), red (B4), NIR (B5), short-wave infrared 1 (SWIR1) (B6), and short-wave infrared 2 (SWIR2) (B7). Landsat 8-OLI's SR data boast a spatial resolution of 30 m. To ensure precision in generating samples, we utilized Sentinel-2 satellite data, which offer a higher spatial resolution of 2 m, alongside Google Earth images, recognized for their close resemblance to real land cover [52]. While Landsat 8 spectrum bands were used as input features, Sentinel-2 and Google Earth images served in the sample generation process.

Furthermore, various data types are required to calculate input features related to topography, vegetation, climate, stream discharge, and social economy, sourced from different outlets. The GDEMV2 digital elevation model (DEM) with

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a 30-m resolution was obtained from the Geospatial Data Cloud, facilitating the derivation of elevation, slope, and aspect values. These topographic features reflect changes associated with land cover types. MOD13Q1 NDVI [53], MOD11A2 land surface temperature (LST) [54], and MCD43A3 albedo [55] data were obtained from the Moderate Resolution Imaging Spectroradiometer (MODIS), offering spatial resolutions of 250, 1000, and 500 m, respectively, and the temporal resolution of 16 days, eight days, and daily. To ensure consistency, these data types were resampled to a unified resolution of 16 days and 30 m. Meteorological daily data for 2020, encompassing daily precipitation, air temperature, wind speed, relative humidity, and solar radiation, were collected from the China Meteorological Data Network and interpolated to a 30-m spatial grid using inverse distance weighting (IDW).

The aforementioned datasets were employed in calculating actual evapotranspiration, with the results necessitating validation alongside terrestrial water storage (TWS) and runoff data. TWS anomalies (TWSAs) are integral components of the hydrological cycle, featuring a spatial resolution of 0.25° and sourced from the Center for Space Research, The University of Texas [56]. Spatial weighting methods were applied to convert TWSA data to a basin scale. Monthly observed runoff data from 44 outlet control stations in subbasins within the Ordos Basin, recorded in 2020, were procured from the Yellow River Conservancy Commission of the Ministry of Water Resources. Three evapotranspiration products were downloaded to validate the simulated results, including GLEAM, MOD16A2, and PML-v2 [57], [58]. In addition, night light data, depicting the distribution and aggregation of urban areas and towns, were derived from the National Oceanic and Atmospheric Administration (NOAA) at a resolution of 500 m [59]. To align with the study's requirements, these data were resampled to a 30-m resolution and employed as one of the input features.

To assess and compare variances, six types of land cover data were employed, including CNLUCC [60], MCD12Q1 [61], Globe Land [24], CLCD [27], GLC_FCS [29], and CRLC [3]. CNLUCC is a widely used product in China, characterized by artificial visual interpretation using Landsat remote sensing imagery as the primary source, boasting an accuracy exceeding 90%. It offers a coarse resolution of 1 km. MCD12Q1 is a global land cover product relying on MODIS satellite data and supervised classification methods, featuring a resolution of 500 m. Globe Land, a global land cover dataset, provides spatial data at a resolution of 30 m and was supplied by China to the United Nations. It covers three years (2000, 2010, and 2020) with an overall accuracy (OA) of approximately 83.5%. CLCD, a Chinese land cover product, offers a resolution of 30 m and was developed by Wuhan University using Landsat data, boasting an accuracy of about 79.3%. GLC_FCS, another global classification product, provides 30-m resolution data for five-year intervals, developed by the Chinese Academy of Sciences, with an OA of approximately 82.5%. CRLC, a fine classification product for China at 10-m resolution, was created by Wuhan University based on 30-m resolution data, achieving an OA of about 84.9%. While these products offer high accuracy, their performance varies across different regions. Further details regarding the datasets used in this study are provided in Table I.

III. Methodology

A. Framework of Land Cover Mapping

This study aimed to introduce a novel method for land cover mapping, encompassing several key steps in its processing chain, including sample selection, actual evapotranspiration estimation, construction of input features and classification system, accuracy assessment, and intercomparison with existing products (see Fig. 2).

Initially, we selected samples using stratified random sampling and visual interpretation methods based on Sentinel-2 imagery from the GEE platform [62]. Google Earth Images were also used to assist in the classification process. Following the classification system, the samples were selected and divided into training and testing sets.

Next, we utilized long time series remote sensing datasets along with the SEBAL model to estimate real-time evapotranspiration during satellite transits. Calculation results were obtained and validated at 16-day intervals. Subsequently, we compiled several input features, encompassing multiband spectral data, spectral indices calculated from Landsat imagery, topographic information, socioeconomic indicators, and hydrothermal conditions. These features were combined to create a comprehensive feature set for classification.

Finally, a classifier was employed to simulate a land cover map, and the map's accuracy was validated by comparing it with data from multiple sources with varying spatial resolutions for quality assessment.

1) Selection of Input Features: The classification features in this study were derived from various factors, including spectrum, topography, social economic data, and hydrothermal process. To establish basic classification features, we utilized all available SR data for the year 2020, specifically the six spectral bands from B2 to B7. In addition, three spectral indices were computed to enhance distinctions among land cover types, including the normalized difference vegetation index (NDVI), the modified normalized difference water index (MNDWI), and the normalized difference built-up index (NDBI) [19]. These spectral indices primarily served to differentiate between vegetation, water bodies, and built-up areas. Furthermore, we found that incorporating topographical factors and light indices as additional information sources improved classification efficiency. Therefore, two topographical factors (DEM and slope) and one socioeconomic indicator (night light data) were also included as input features.

While NDVI was used to distinguish vegetation from nonvegetation, it did not adequately differentiate between different vegetation types, such as forests, grasslands, and croplands. To address this, 21-day instantaneous evapotranspiration data are added to classification features for the entire year 2020. It is worth noting that there is a total of 23 days with instantaneous evapotranspiration in 2020 (every 16 days), with calculations not available on April 6 and August 28 due to meteorological reasons. Thus, only 21 days of ET_{inst} data were effectively used for classification. In summary, we obtained a

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TABLE I
DATASET LIST AND DETAILED INFORMATION

Data	Resolution	Data sources
Landsat 8	30m	GEE [a]
Sentinel-2	2m	GEE
Google Earth	0.5m	GEE
GDEMV2 DEM	30m	Geospatial Data Cloud [b]
MOD13Q1 NDVI	16day&250m	MODIS [c]
MOD11A2 LST	8day&1000m	MODIS
MCD43A3 Albedo	Daily&500m	MODIS
Meteorological data V3.0	Daily&Station	China Meteorological Data Network [d]
Runoff	Monthly&Station	Water year book
MOD16A2 ET	8day&500m	MODIS
GLEAM ET	Daily&0.25°	[57]
PML-v2 ET	8day&0.05°	[58]
VIIRS Night light	Monthly&500m	NOAA [e]
GRACE Terrestrial Water Storage	Monthly&0.25°	Center for Space Research [f]
CNLUCC	1km	RESDC [g]
MCD12Q1	500m	MODIS
Globe Land	30m	GLOBELAND30 [h]
CLCD	30m	[27]
GLC_FCS	30m	[29]
CRLC	10m	[3]

[a] Google Earth Engine Platform (GEE), https://earthengine.google.com/
[b] Geospatial Data Cloud, http://www.gscloud.cn/
[c] MODIS, https://ladsweb.modaps.eosdis.nasa.gov/
[d] China Meteorological Data Network, http://data.cma.cn
[e] NOAA, https://ngdc.noaa.gov/eog/dmsp/download_radcal.html
[f] Center for Space Research, https://www2.csr.utexas.edu/grace/RL06_mascons.html
[g] Resource and Environment Science and Data Center (RESDC), http://www.resdc.cn
[h] GLOBELAND30, http://www.globallandcover.com

total of 33 features, comprising six spectral bands, three normalized spectral indices, 21 ET_{inst} data points, two topographic features, and one social–economic feature.

To compare the accuracy of different input features, we defined five input schemes in this study. Scheme 1 (S1) used only the six spectral bands as input features. Scheme 2 (S2) included the six spectral bands and three spectral indices for classification. Scheme 3 (S3) added elevation and slope information to the features used in S2. Scheme 4 (S4) encompassed spectral bands, spectral indices, DEM, and night light data for classifier construction. Scheme 5 (S5) included all input data, comprising six spectral bands, three normalized spectral indices, 21 ET_{inst} data points, two topographic features, and night light data.

Furthermore, we conducted variance analysis among input features for different land cover types to compare the contribution of each input feature to classification. When the difference in input features among different land cover types is greater, the contribution of that input feature to classification is considered higher. We employed two statistical indicators, Kullback–Leibler divergence (KLD) and Jensen–Shannon divergence (JSD), to measure the differences in different feature inputs among land cover types.

KLD can represent the difference between two probability distributions. For continuous random variables, suppose that there are two probability distributions F and P on the probability space X, and their probability densities are f and p, respectively. The KL divergence of F for P is defined as follows:

$$\mathrm{KL}(F, P) = \int_{-\infty}^{+\infty} \ln \frac{f(x)}{p(x)} dx. \tag{1}$$

For discrete random variables, then the KL divergence of F for P is defined as follows:

$$D(F\|P) = \sum_{i} F(x) \ln \frac{F(x)}{P(x)} = \sum_{x \in X} F(x)(\ln(F(x) - \ln(P(x))). \tag{2}$$

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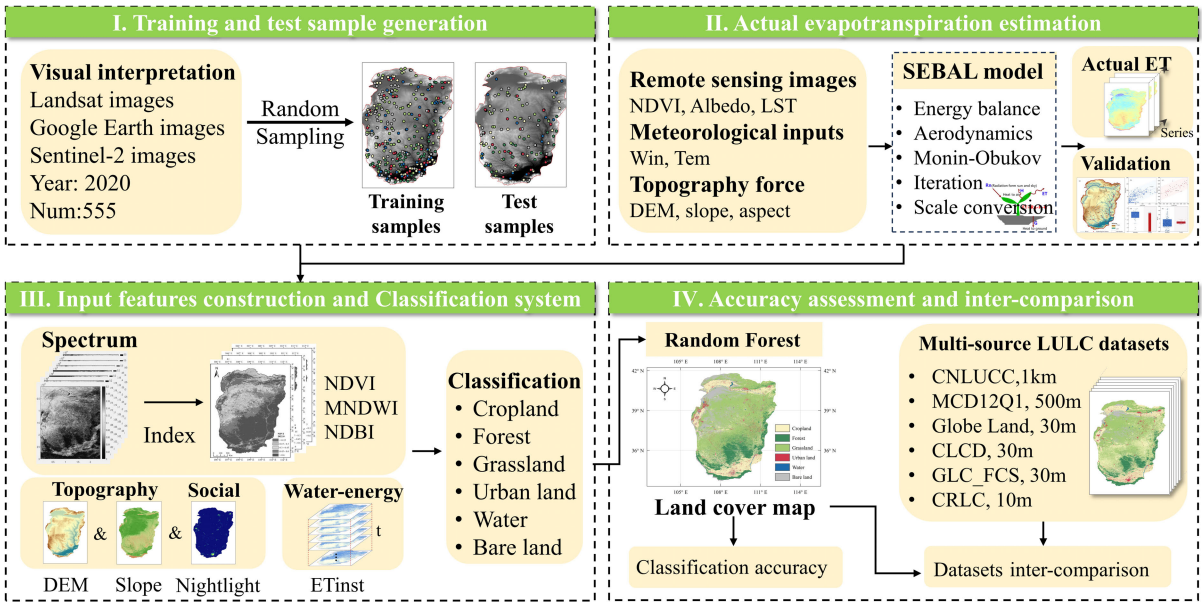


Fig. 2. Framework to generate the land cover map of Ordos Basin.

According to the definition of KLD, it is asymmetric. It can be transformed into a symmetric JS divergence by a slight modification

$$\mathrm{JS}(F, P) = \frac{1}{2}[D(F\|P) + D(P\|F)]. \tag{3}$$

Similarly, the ranges of KL and JS are both [0, 1]. When the two probability distributions are exactly the same, KL or JS is 0, and KL or JS is 1 conversely. In other words, the larger the value of KL or JS, the higher the difference between the two probability distributions, that is, the higher the difference between the variables.

2) Classification System: This study employs a classification system comprising six land cover types: cropland, forest, grassland, urban land, water, and bare land. A total of 555 samples were selected using random sampling and visual interpretation techniques, primarily based on Sentinel-2 remote sensing imagery, which provides relatively high resolution. These samples were randomly divided into 388 training samples and 167 testing samples by stratified randomization.

The random forest (RF) classifier was chosen for land cover mapping due to its ability to handle high-dimensional data, tolerance to sample errors, and robustness to missing data [63], [64]. RF consistently demonstrates high prediction accuracy compared to other statistical models [65]. Therefore, an RF classifier was used to generate the land cover map. The land cover map obtained through the proposed method in this article is referred to as land use and land cover mapping using evapotranspiration (ETLULC).

3) Accuracy Assessment: Four indexes are used to evaluate the result of land cover classification in this article based on the confusion matrixes, including OA, producer's accuracy (PA), user's accuracy (UA), and $F1$ score conveys ($F1$). OA means the proportion of correctly classified pixels in the total pixel, which directly reflects the proportion of correctly classified pixels. PA and UA calculated the accuracy of one of the land cover types, respectively. The former is the ratio of the number of pixels that the classifier correctly classifies into land cover classes and the total number of true land cover classes in the entire image. The latter is the ratio of the total number of pixels correctly assigned to the land cover class and assigned to the land cover class by the classifier in the image. The $F1$ score conveys the balance between PA and UA and is calculated as

$$F1 = 2\frac{\mathrm{PA} \times \mathrm{UA}}{(\mathrm{PA} + \mathrm{UA})} \times 100\%. \tag{4}$$

The map accuracy is usually evaluated by samples; it is imperfect to directly compare. The variances of the estimates should be taken into account for accuracy comparison [66]. In order to compare the accuracy of different input schemes in contrast experiments, statistical variance index z is used to test the variance [67]

$$z = \frac{\left|\frac{X_1}{N_1} - \frac{X_2}{N_2}\right| - \frac{1}{2}\left(\frac{1}{N_1} + \frac{1}{N_2}\right)}{\sqrt{\omega(1-\omega)\left(\frac{1}{N_1} + \frac{1}{N_2}\right)}} \tag{5}$$

where X_1 and X_2 represent the number of correctly allocated cases in two independent samples of size N_1 and N_2, and $\omega = (X_1 + X_2)/(N_1 + N_2)$. If there is a difference between the two results, the null hypothesis of no significant difference would be rejected if $|z| > 1.96$ at a 5% level of significance.

To do the datasets' intercomparison with the existing land cover products, six types of land cover data were used to assess and compare accuracy, including CNLUCC (1 km), MCD12Q1 (500 m), Globe Land (30 m), CLCD (30 m), GLC_FCS (30 m), and CRLC (10 m). The LULC datasets in different resolutions are recognized as widely accepted data. The proposed ETLULC is focused on the study area, while the other products are trained on different algorithms and more extensive data for a global or national assessment. The similarities and discrepancies are tested based on the comparison framework established by Duarte et al. [68] to

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ensure a fairness comparison with other products. The overall proportion of agreement (OPA) and the marginal proportions of agreement (MPA) per class among maps are calculated by the following formulas:

$$\text{OPA} = \frac{\sum_{i=1}^{n} p_{ii}}{\text{Total number of pixels}} \tag{6}$$

$$\text{MPA}_{\text{ETLULC-PRODUCT}}(i) = \frac{\sum_{i=1}^{n} p_{ii}}{\text{Number of pixels of class } i \in \text{ETLULC}} \tag{7}$$

$$\text{MPA}_{\text{PRODUCT-ETLULC}}(i) = \frac{\sum_{i=1}^{n} p_{ii}}{\text{Number of pixels of class } i \in \text{PRODUCT}} \tag{8}$$

where p_{ii} is the number of pixels classified with the same class in ETLULC and products, and n is the number of classification types. In fact, $\text{MPA}_{\text{ETLULC-PRODUCT}}(i)$ and $\text{MPA}_{\text{PRODUCT-ETLULC}}(i)$ correspond to the PA and UA when considering one of the maps as the reference.

B. Actual Evapotranspiration Estimation Based on Remote SEBAL Model

SEBAL is a remote sensing model to estimate the regional real evapotranspiration [69], [70]. The SEBAL model is based on spectral bands of remote sensing images and meteorological data to invert regional surface feature parameters and then calculate the instantaneous net radiation flux (R_n), soil heat flux sensible (G), and heat flux (H) during satellite transit. Actual instantaneous evapotranspiration can be calculated by the principle of energy balance, which can also be converted to actual evapotranspiration of a day or period.

1) Calculating Procedure of SEBAL Model: The basic theory of the SEBAL model is the energy balance equation given as follows:

$$\text{LE} = R_n - G - H \tag{9}$$

where LE is latent heat flux.

R_n refers to the energy obtained by the surface under solar radiation. The calculation formula is given as follows:

$$R_n = (1-\alpha)K_{\text{in}} + (L_{\text{in}} - L_{\text{out}}) - (1-e)L_{\text{in}} \tag{10}$$

where α is land surface albedo, L_{in} is incident long-wave radiation, e refers to emissivity, L_{out} is surface-reflected long-wave radiation, and K_{in} is incident shortwave radiation.

G refers to soil heat exchange capacity per unit of time area and is mainly affected by soil thermal conductivity and temperature gradient between surface height and reference height. The temperature gradient cannot be obtained from remote sensing data, so it is calculated by empirical formula in the SEBAL model

$$G = \frac{T_S - 273.16}{\alpha}(0.0038\alpha + 0.0074\alpha^2) \cdot (1 - 0.98\text{NDVI}^4)R_n \tag{11}$$

where T_s is LST.

H is the energy transferred from the surface to the air through heat exchange. The calculation of H requires aerodynamic iteration and Monin–Obukov correction

$$H = \frac{\rho_{\text{air}} \cdot C_P \cdot dT}{r_{\text{ah}}} \tag{12}$$

where ρ_{air} is the air density, C_P is the air-specific heat at constant pressure, r_{ah} refers to the aerodynamic impedance of heat transfer, and dT refers to the temperature difference at heights z_1 and z_2 (T_1–T_2) above zero plane displacement.

LE means the energy required for the evaporation of ground water into the air, which determines the value of evapotranspiration. The instantaneous LE at the time of satellite transit can be calculated by formula (1).

After calculating R_n, G, H, and LE, four components of energy balance, instantaneous evapotranspiration (ET_{inst}) at the time of image transit can be calculated using the following formula:

$$\text{ET}_{\text{inst}} = 3600 \cdot \text{LE} \cdot \cos(s)/\lambda \tag{13}$$

where λ refers to the latent heat of vaporization and s is the cosine of the surface slope.

2) Validating Procedure of SEBAL Results: It is difficult to verify the simulation results of instantaneous evaporation due to the lack of observed data at the hour scale. The scale expansion method can be used to convert instantaneous evaporation into cumulative evaporation over time periods for accuracy validation. The method is given as follows. Assuming that during the energy balance process, the relative proportion of each flux component is stable during the day time, and ignoring the soil heat flux that offsets each other during the day and night, the daily evaporation ratio EF is assumed to be a constant

$$\text{EF} = \frac{\text{LE}}{Rn - G} = \frac{Rn - G - H}{Rn - G}. \tag{14}$$

Daily actual evapotranspiration ET_{24} can be estimated as follows:

$$\text{ET}_{24} = \frac{86400 \cdot \text{EF} \cdot (R_{n24} - G_{24})}{l} \tag{15}$$

where R_{n24} and G_{24} refer to the daily net radiation flux and the daily average soil heat flux, respectively.

There is a problem that daily satellite images are not available every day by using remote sensing data; it is possible to extend the image day as the center to a certain date. Thus, the cumulative ET calculation for a certain period of time is given as follows:

$$\text{ET}_{\Delta t} = \sum_{i=m}^{n} [(\text{ET}_{24r} P_i)(\text{ET}_{24ri})] \tag{16}$$

where $\text{ET}_{\Delta t}$ is the total evaporation from day m to day n, $\text{ET}_{24r} P_i$ is the reference evapotranspiration ratio on day i, and ET_{24ri} is the reference evapotranspiration on day i.

As for a closed basin, the water balance formula can be written as

$$P = R + \text{ET} + \Delta S \tag{17}$$

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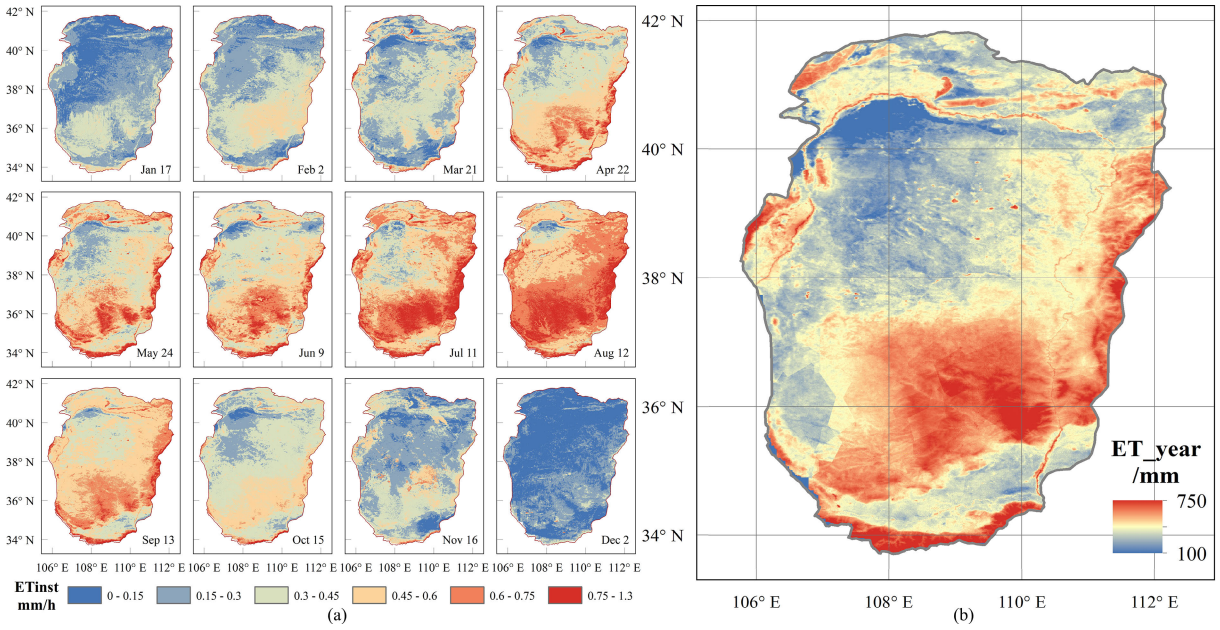


Fig. 3. Multiperiod instantaneous evapotranspiration and annual evapotranspiration estimated by SEBAL model in 2020. (a) Multiperiod instantaneous evapotranspiration. (b) Annual evapotranspiration.

where P is the precipitation, R is the runoff, ET is the actual evapotranspiration, and ΔS is the change of water storage capacity in the basin.

The Ordos Basin was divided into 44 subbasins according to the hydrological control station. Then, the actual evapotranspiration was calculated as the real evapotranspiration (E_{obs}) based on the water balance formula. The result of the SEBAL model (E_{sim}) was compared with E_{obs}, and the correlation coefficient r was calculated later

$$r = \left(\sum_{i=1}^{n} \left[\left(E_{\text{obs}} - \overline{E} \right) \left(E_{\text{sim}} - \overline{E_s} \right) \right]^2 \Big/ \sum_{i=1}^{n} \left(E_{\text{obs}} - \overline{E} \right)^2 \sum_{i=1}^{n} \left(E_{\text{sim}} - \overline{E_s} \right)^2 \right)^{\frac{1}{2}} \tag{18}$$

where E_{obs} and E_{sim} are real evapotranspiration and simulated evapotranspiration and $\bar{E}$ and $\bar{E}_s$ are the mean values of real and simulated evapotranspiration, respectively. The closer r approaches 1, the closer the simulated value is to the real value, and the better the simulation performance is. Moreover, three products were applied to validate the simulated results, including GLEAM, MOD16A2, and PML-v2 at a monthly scale.

IV. Results

A. Evapotranspiration Calculation and Validation at a Large Spatial Scale

The actual evapotranspiration for Ordos Basin in 2020 was estimated using the SEBAL remote sensing evapotranspiration model. SEBAL requires several input parameters, including NDVI, albedo, LST, and meteorological data, with datasets available at approximately 16-day intervals throughout the year. Over the entire year, 21 days of instantaneous evapotranspiration (ET_{inst}) were simulated during the satellite's transit.

Fig. 3 presents the spatial distribution of ET_{inst} on a monthly basis throughout the year. Notably, ET_{inst} exhibits a pattern of initial increase followed by a decrease over time, with significantly higher evapotranspiration occurring from April to September. The unit of ET_{inst} is mm/hr, with mean values ranging from below 0.4 mm from January to March to approximately 0.5–0.7 mm from April to September, reaching over 1 mm in certain areas. The temporal variation in evapotranspiration can be attributed to factors such as temperature and precipitation, with peak values occurring during the rainy season. It is primarily concentrated in spring and summer, with over 65% occurring from April to September, which is also similar to the studies of Lian and Huang [71] and Gao et al. [72].

Spatially, ET_{inst} generally decreases from the southeast to the northwest of the Ordos Basin. The southern loess area exhibits higher evapotranspiration compared to the northern sandy area. In addition, the spatial distribution of ET_{inst} aligns with prior knowledge of land cover, indicating that forests generally have higher evapotranspiration than croplands and grasslands. Cropland and grassland exhibit relatively similar evapotranspiration rates, with cropland slightly higher. This observation reflects the spatial distribution differences in vegetation types. The distribution of evapotranspiration is similar to Yang's study [73].

To validate the SEBAL estimation results, the scale of ET_{inst} was expanded to obtain monthly and annual evapotranspiration values, which were verified using the water balance method for the Ordos Basin. As shown in Fig. 4(a), the basin was divided into 44 subbasins to represent different characteristics. The correlation coefficient (r) between observed evapotranspiration (E_{obs}) and simulated evapotranspiration (E_{sim}) on a monthly scale was 0.63 for all subbasins, while, at the annual scale, it increased to 0.78 [see Fig. 4(b) and (c)]. Most subbasins exhibited an r value above 0.6 on the monthly scale, with

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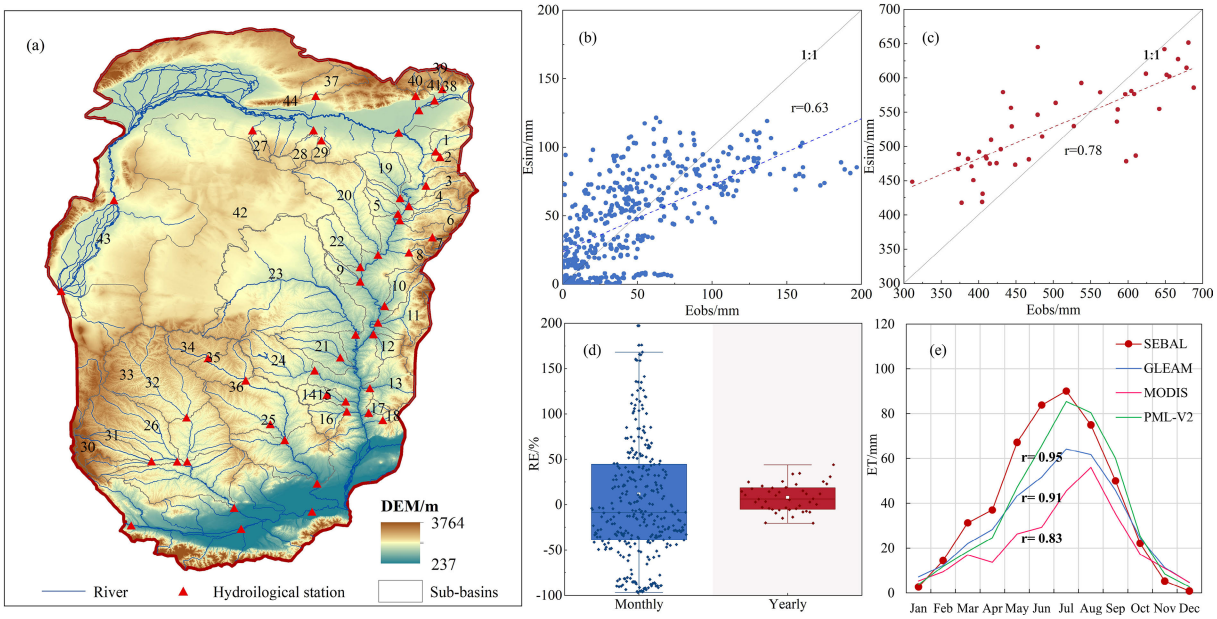


Fig. 4. Validation of SEBAL results in subbasins. (a) Location of 44 subbasins and hydrological-control station. (b) Scatter plot of E_{sim} and E_{obs} on the monthly scale. (c) Scatter plot of E_{sim} and E_{obs} on the yearly scale. (d) Relative error between simulated and water-balance evapotranspiration in subbasins. (e) Annual variation of simulated results and products.

an average monthly relative error of approximately 11.5%. However, some monthly errors were relatively large, potentially due to errors in large-scale TWS data when applied to smaller basins. At the monthly scale, the annual variation was compared between simulated results and three products [see Fig. 4(e)]. There are differences in the comparison results of products. The correlation coefficients among GLEAM, MODIS, PML-V2, and SEBAL simulation are both above 0.8, indicating good overall consistency between the simulation results and existing products. At the annual scale, the evapotranspiration in 2020 generally falls within the range of 100–750 mm in the Ordos Basin. The relative error was mostly within 10%, demonstrating good agreement between actual evapotranspiration and simulated evapotranspiration. The MEP and SEBAL models were applied to estimate the actual evapotranspiration in the Loess Plateau in [72] and [73]. The annual evapotranspiration ranges from 100 to 660 mm. Ordos Basin is located in the core area of the Loess Plateau with similar geographical attributes. It confirmed the reliability of the SEBAL model's simulated evapotranspiration. Moreover, the results also align with previous studies' findings [74], [75].

B. Result and Accuracy Assessment of Land Cover Mapping With Evapotranspiration

1) Classification Result and Accuracy Assessment: The land cover map of Ordos Basin, referred to as ETLULC, is presented in Fig. 5. Overall, the accuracy of ETLULC was consistently high and satisfactory, with an OA of 93.1%. The accuracy of the six land cover types in ETLULC was assessed using visually interpreted independent samples. A total of 555 samples from the Ordos Basin were divided into training and testing samples. In the training samples, all land cover types exhibited high accuracy, with PA, UA, and $F1$ scores all exceeding 90%, resulting in an OA close to 1. In the test samples, the OA was 80.7%. While the PA and UA of cropland were both 78.1%, the PA, UA, and $F1$ scores of forest were 89.7%, 83.3%, and 86.4%, respectively. For grassland, all metrics were approximately 75%. Similarly, urban, water, and bare land had $F1$ scores above 75%. In terms of accuracy across all samples, the OA was 93.1%, with PA and UA for all land cover types exceeding 85% and $F1$ scores exceeding 90%. In summary, the classification results were generally satisfactory.

The proportions of the six land cover types in the Ordos Basin are illustrated in the pie chart at the bottom left of Fig. 5. Grassland had the largest proportion (41%), followed by cropland (22.3%) and forest (17.7%). Bare land accounted for 14.2%, primarily consisting of desert or sandy soil. Urban land and water covered about 3.6% and less than 1%, respectively. In total, the three vegetation types (cropland, forest, and grassland) dominated the Ordos Basin's land cover, accounting for more than 81%. The spatial distribution of land cover in the Ordos Basin revealed three concentrated cropland zones: the Weinan Plain in the south, the Hetao Irrigation District in the north, and the Ningxia Irrigation District in the northwest, all significant agricultural regions. Forests were primarily distributed in the southeastern mountainous regions of the Loess Plateau. Grassland exhibited the broadest distribution, mainly in the northern sandy areas (interspersed with desert or bare land) and transitional zones between sandy and loess areas (interspersed with cropland). Water bodies were primarily aligned with the main streams and tributaries of the Yellow River, contributing to agricultural development. Urban land distribution closely mirrored areas of concentrated cropland and rivers. Bare land in the Ordos Basin consisted mostly of deserts or sandy soil, such as the Kubuqi Desert and Mu Us Sandland. This flat terrain was covered with windblown sand, featuring a mixed distribution

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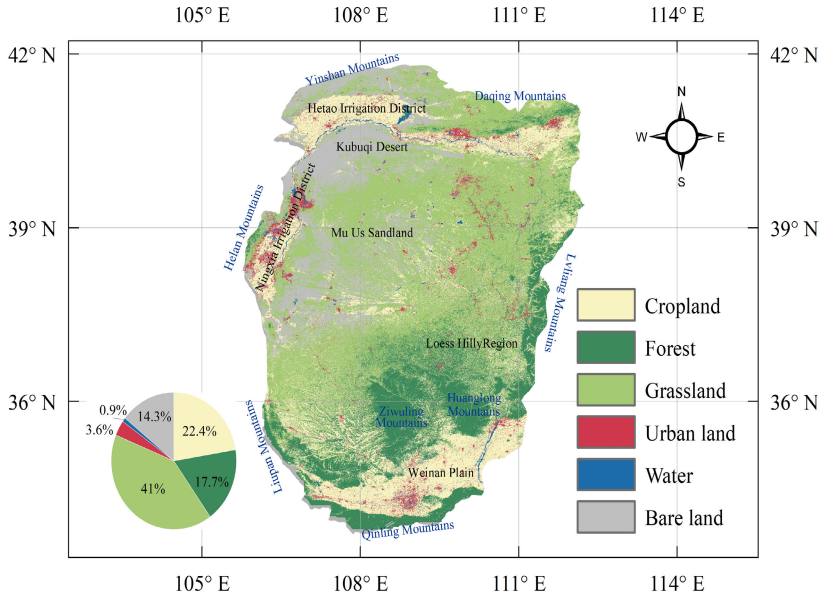


Fig. 5. Land cover mapping of Ordos Basin in 2020 (the pie chart at the bottom left represents the proportions of the six land cover types).

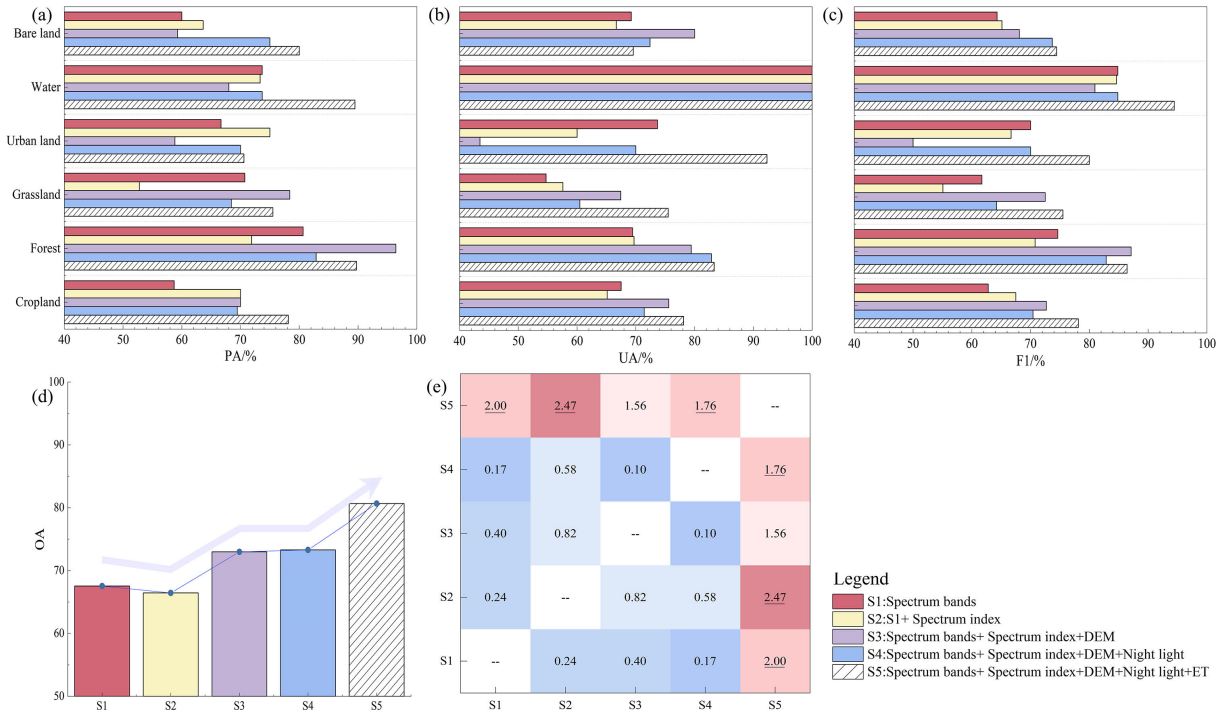


Fig. 6. Accuracy and variance comparison among different combinatorial input schemes. (a) PA. (b) UA. (c) F1. (d) OA. (e) Variance index z.

of shrubs and grassland due to its topographical characteristics. The classification results aligned well with the actual land cover distribution in the Ordos Basin [3], [29].

2) Accuracy and Variance Comparison Under Different Input Schemes: Different input schemes, consisting of various input combinations, were compared to investigate the impact of adding instantaneous evaporation as an additional feature in comparison to traditional classification methods (see Fig. 6). Five input schemes were compared using testing samples: S1 (spectrum bands only), S2 (spectrum bands and spectrum index), S3 (spectrum bands, spectrum index, and DEM), S4 (spectrum bands, spectrum index, DEM, and night light), and S5 (spectrum bands, spectrum index, DEM, night light, and evapotranspiration). The ETLULC mapping was generated using the fifth input scheme.

The variance in PA, UA, and $F1$ scores among the five different input schemes is presented in Fig. 6(a)–(c). It is important to note that accuracy comparisons were conducted using testing samples to ensure fairness; these samples were not part of the training dataset used for the RF classifier and served as independent validation samples. When only spectral bands were used as classification features (S1), the

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overall classification accuracy was around 60%–70%. However, the accuracy was relatively low for cropland (62.8%) and grassland (61.7%). When both spectral bands and spectral indexes were used for classification (S2), the $F1$ score for cropland increased by 4.7%, while the $F1$ scores for other land cover types either did not increase or even decreased slightly. The OA of S2 was similar to that of S1, indicating that NDVI, NDBI, and MNDWI contributed insignificantly to performance. When DEM was added to S2 (S3), the $F1$ scores for cropland, forest, and grassland increased by 5.2%, 16.33%, and 17.43%, respectively, demonstrating the significant role of topographical factors in improving differentiation among different vegetation types. Cropland is typically associated with specific slope conditions, while forested land is often found at higher elevations [76], [77]. S4, which incorporated spectral bands, spectral indexes, DEM, and night light, notably improved the classification accuracy of urban land, increasing it by 0.1%, 3.3%, and 20%, respectively, compared to the first three input schemes. Night light played a significant role in identifying urban areas [59]. The final input scheme (S5) included all the aforementioned features as inputs for the RF classifier. Compared to the previous input schemes (S1–S4), S5 resulted in an OA improvement of 7.4%–14.2%. For the six land cover types, $F1$ scores changed by 5.4%–15.3%, −0.7%–15.7%, 3.0%–20.4%, 9.9%–29.9%, 9.6%–13.5%, and 0.7%–10.1%, respectively.

Grassland, urban land, and bare land showed the most significant accuracy improvements in classification when considering evapotranspiration (ETLULC). For the three vegetation types (cropland, forest, and grassland), the forest consistently exhibited relatively high accuracy, indicating that spectral information for forests is more distinguishable compared to cropland and grassland. The $F1$ score for forest in S5 improved by about 11.8%, 15.7%, −0.7%, and 3.6% compared to S1–S4. Grassland accuracy improved by 13.8%, 20.4%, 3.0%, and 11.3% compared to S1–S4. Cropland accuracy also improved by 15.3%, 10.7%, 5.4%, and 7.7%, respectively. Regarding OA, S1 (67.6%) was similar to S2 (66.5%), while S3 (73.0%) was akin to S4 (73.3%). This suggests that adding spectral indexes on top of S1 or including night light based on S3 had little impact on the results. Spectral indexes are derived from spectral bands and do not introduce new external information for classification. Night light, while aiding urban area identification, had a limited effect due to the relatively small proportion of urban land. Notably, S3 saw significant improvement in accuracy upon adding DEM, underscoring the substantial role of elevation and slope in vegetation classification, as discussed earlier. Significantly, S5 demonstrated marked improvement after incorporating ET_{inst}, further confirming the effectiveness of instantaneous evapotranspiration as an input feature. Among the three vegetation types, the classifier in S5 exhibited the greatest accuracy improvement in grassland classification, followed by cropland. In addition, the variance between the two input schemes is depicted in Fig. 6(e). Variance among S1–S4 was relatively small, with statistical indicators (z) ranging from 0.1 to 0.9, failing to pass significance tests. Conversely, the variance between S5 and S1–S4 was significant, with statistical indicators (z) ranging from 1.6 to 2.5. The underlined numbers indicate significance at a 5% level. This indicates that considering differences in evapotranspiration among various vegetation types can improve vegetation classification accuracy to some extent, particularly for two easily confused land cover types: cropland and grassland. Detailed explanations are provided in Section V. This approach holds promise in regions with mixed vegetation or similar vegetation spectra when utilizing the ETLULC framework.

C. Intercomparison With Existing Thematic Products

To facilitate a qualitative comparison with existing land cover products, we have compiled spatial maps of multisource LULC data, encompassing six distinct land cover products and the classification results presented in this study, as depicted in Fig. 7. These products comprise CNLUCC, CLCD, CRLC, MCD12Q1, Globe Land, and GLC-FCS. CNLUCC, CLCD, and CRLC represent products at the national scale for China, while MCD12Q1, Globe Land, and GLC-FCS provide global-scale coverage. It is worth noting that these datasets possess varying spatial resolutions of 1 km and 500, 30, 30, 30, and 10 m, respectively. In contrast, the land cover mapping developed for the Ordos Basin in this study (ETLULC) is at a 30-m spatial resolution.

Upon analyzing the spatial distribution of land cover types across these datasets, it is evident that the general distribution patterns of land cover features are relatively similar, albeit with notable differences in specific details. Among these products, CLCD, GLC-FCS, and CRLC exhibit closer similarities to ETLULC within the Ordos Basin. In the CNLUCC dataset, land cover appears more dispersed, leading to the possibility of multiple land cover types coexisting within a single grid cell due to its 1-km spatial resolution. This may result in substantial errors when assigning a dominant land cover type to such grid cells, potentially overlooking the presence of other land cover types within the same grid. In contrast, the MCD12Q1 product, a global MODIS-based dataset, exhibits relatively concentrated land cover types, particularly in grassland and cropland, which do not adequately represent the true complexity of vegetation mixing and significantly differ from the classification results. These findings underscore the limitations of applying large-scale and coarse-resolution remote sensing products to regional assessments. While Globe Land, MCD12Q1, and GLC-FCS provide land cover classifications at 30-m resolution, CRLC offers a finer 10-m resolution. At this increased spatial resolution, land cover distribution largely aligns, especially concerning forests. However, slight disparities persist in the classification of cropland, grassland, and bare land. To further evaluate the spatial details of different maps and our proposed ETLULC, we randomly selected three small regions for assessment, as illustrated in Fig. 7(a)–(c). The results demonstrate a favorable agreement between ETLULC and the Sentinel-2 image landscape.

To comprehensively assess the congruence and disparities between the classification results (ETLULC) and existing thematic products, we calculated OPA and MPA in comparison to these products, yielding the results depicted in Fig. 8 and Table II. ETLULC consistently exhibited strong consistency

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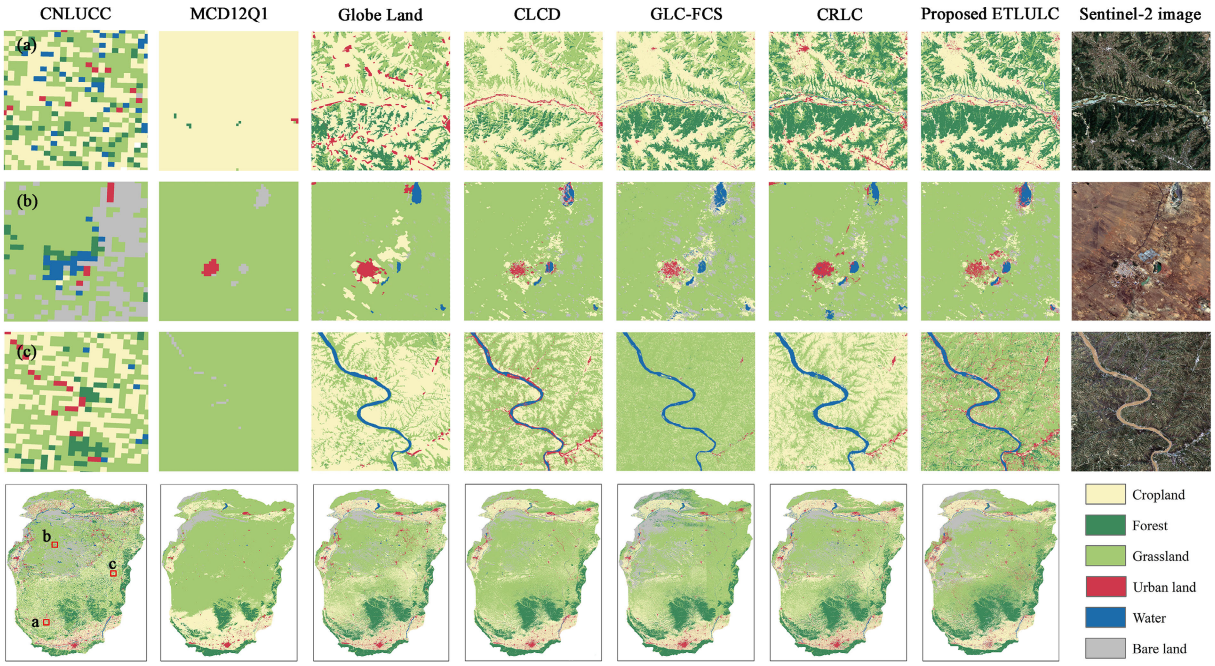


Fig. 7. Comparison among the land cover mapping obtained by the proposed ETLULC framework and the other land cover products. (a)–(c) Partial enlarged detail of randomly selected pixels.

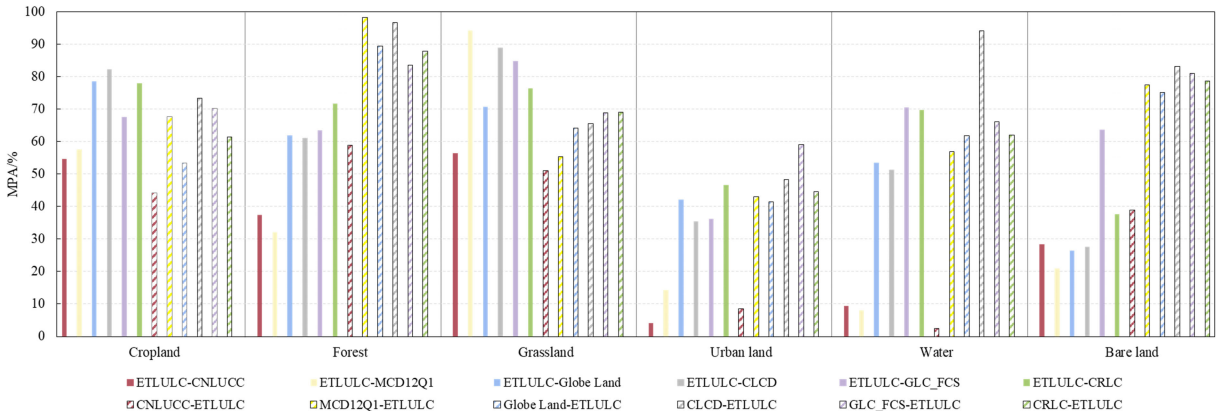


Fig. 8. Comparison of MPAETLULC-PRODUCT and MPAPRODUCT-ETLULC for multisource LULC datasets for each land cover type.

TABLE II
OPA FOR PROPOSED ETLULC AND SIX LULC PRODUCTS

%	CNLUCC	MCD12Q1	Globe Land	CLCD	GLC_FCS	CRLC
OPA	46.3	60.9	63.4	71.5	72.3	69.3

with existing land cover products, affirming the reliability of our classification results. The highest similarity was observed between ETLULC and GLC-FCS (72.3%), followed by CLCD (71.5%), CRLC (69.3%), Globe Land (63.4%), MCD12Q1 (60.9%), and CNLUCC (46.3%).

The MPA indicates the accuracy or similarity of land cover classes in ETLULC compared with other products (represented by 100%—$MPA_{PRODUCT\text{-}ETLULC}$) and vice versa ($MPA_{ETLULC\text{-}PRODUCT}$) and further underscored the strong agreement between ETLULC and CLCD, GLC-FCS, and CRLC. For instance, $MPA_{ETLULC\text{-}GLC_FCS}$ was 84.8%, and $MPA_{GLC_FCS\text{-}ETLULC}$ was 68.9% for grassland. It signifies that 84.8% of the grassland regions in GLC_FCS were correctly classified in ETLULC, while 68.9% of grassland regions in ETLULC were accurately classified in GLC_FCS. In other words, in ETLULC, 15.2% of regions designated as grassland were not similarly classified in GLC_FCS, whereas 31.1% of regions designated as grassland in GLC_FCS were not classified as such in ETLULC.

In terms of specific land cover types, ETLULC demonstrated better agreement with CLCD, GLC-FCS, and CRLC, while the agreement with CNLUCC, MCD12Q1, and Globe Land was relatively lower. For cropland, the consistency between ETLULC and the six products was approximately

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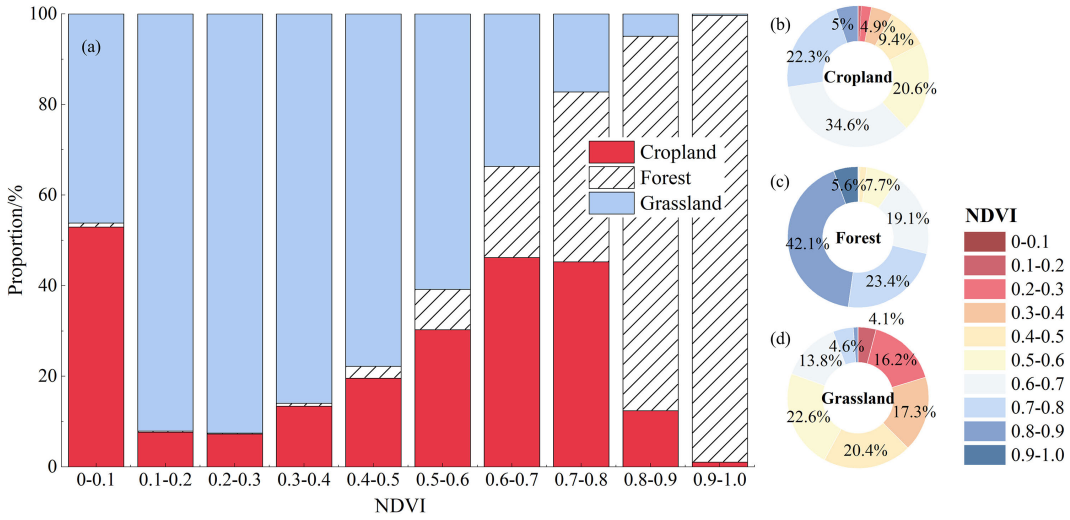


Fig. 9. Proportion of vegetation types in NDVI zones and NDVI distribution in different vegetation types. (a) Proportion of vegetation types in NDVI zones. (b) NDVI distribution in cropland. (c) NDVI distribution in forest. (d) NDVI distribution in grassland.

54.7%, 57.7%, 78.5%, 82.3%, 67.5%, and 78.0%, respectively. For forest, the consistency ranged from 37.4% to 71.7%, while, for grassland, it varied from 56.4% to 94.3%. In contrast, for urban land, water, and bare land, the regions corresponding to consistency ranged from 4% to 47%, 9% to 71%, and 20% to 64%, respectively. Furthermore, there was a gradual improvement in consistency with an increase in the spatial resolution of the product. $MPA_{PRODUCT\text{-}ETLULC}$ consistently exceeded 60%, indicating that the proposed ETLULC captured most of the classification information from various products, especially forest (averaging 85%), bare land (79%), and water (68%). When existing products were applied to the Ordos Basin, overestimation of cropland and underestimation of forest and grassland were observed.

D. Relationship of Land Cover Type and Evapotranspiration

1) Evapotranspiration Characteristics in Different Land Cover Types: In this section, we delve into the variations and dynamics of evapotranspiration across different land cover types, specifically focusing on cropland, forest, and grassland. We base our analysis on the classification results and the computed evaporation data. Given the considerable expanse of the Ordos Basin, we partitioned the region into multiple subregions at various levels, enabling us to scrutinize the interplay and distinctions among land cover types, precipitation, evapotranspiration, and NDVI. The spatial distribution of precipitation and NDVI is depicted in Fig. 1.

The Ordos Basin stretches across a substantial north–south gradient, with 2020 precipitation levels ranging from approximately 200 to 800 mm. Precipitation exhibits a diminishing trend from the southeast to the northwest [see Fig. 1(d)]. Annual evapotranspiration distribution across the Ordos Basin in 2020 is delineated in Fig. 3. It reveals a similar trend of decreasing values from the southeast to the northwest yet highlights regions in the southeast, east, and southern fringes characterized by elevated evapotranspiration values. A comparative examination with the land cover map (see Fig. 5) unveils that regions marked by high evapotranspiration primarily align with forest land cover types, often situated in elevated mountainous or plateau regions. In contrast, central and northern regions exhibit lower evapotranspiration, predominantly featuring a mix of bare land and grassland characterized by limited water resources and reduced evapotranspiration rates. Furthermore, NDVI serves as a proxy for vegetation density, aiding in the discrimination between vegetated and nonvegetated land covers. The spatial distribution of NDVI mirrors the trends observed in precipitation and evapotranspiration, declining from southeast to northwest and closely paralleling the evapotranspiration distribution. The study area encompasses the Ningmeng Irrigation Area, Windblown Sand Area, Loess Plateau Area, and Weihe Plain Area from north to south, each distinguished by unique vegetation characteristics that align with NDVI patterns.

First, the relationship between vegetation types and NDVI is depicted in Fig. 9. NDVI values range from 0 to 1, divided into ten intervals at intervals of 0.1. Fig. 9(a) illustrates the proportion of each vegetation type within each NDVI interval. It is evident that nearly every vegetation type is distributed across the NDVI intervals. The forest is minimally represented when NDVI is below 0.4 but becomes predominant in areas with NDVI exceeding 0.8, suggesting that forested land cover is distinguishable in both low and high NDVI regions. Conversely, within the middle range of NDVI ($0.1 < \text{NDVI} < 0.6$), both cropland and grassland exhibit considerable presence. When NDVI falls below 0.1 and exceeds 0.5, distinguishing between cropland and grassland becomes challenging. This underscores that NDVI primarily distinguishes between vegetated and nonvegetated areas, with limited capacity to differentiate between distinct vegetation types.

Fig. 9(b)–(d) further illustrates the distribution of NDVI within various vegetation types across NDVI intervals. Notably, cropland, forest, and grassland display concentrated NDVI values within specific ranges, often with some overlap, complicating their differentiation. For example, although approximately 80% of forested land exhibits NDVI values in

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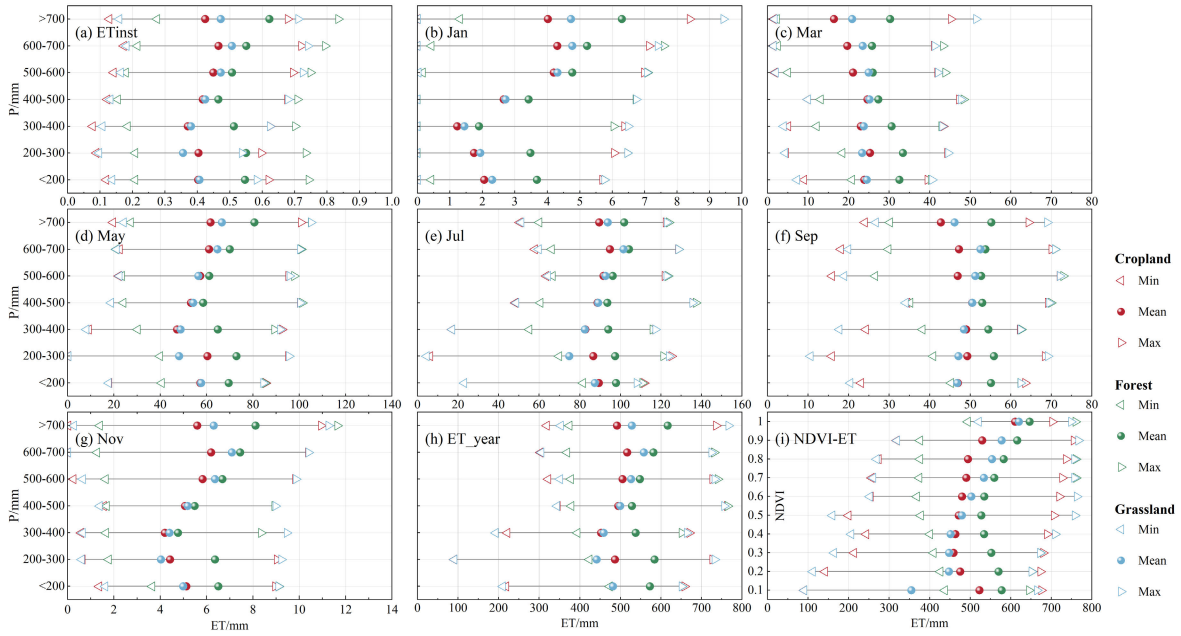


Fig. 10. Comparison of evapotranspiration in different land cover types and rainfall zones. (a) Instantaneous evapotranspiration. (b)–(g) Typical monthly evapotranspiration. (h) Annual evapotranspiration. (i) Comparison of annual evapotranspiration in different NDVI and rainfall zones.

the 0.6–0.9 range, this does not imply that land with NDVI values in this range is exclusively forested. Similarly, 40% and 20% of cropland and grassland, respectively, fall within this NDVI range, underscoring the challenges of distinguishing cropland and grassland based solely on NDVI. Consequently, NDVI alone may require supplementary information for precise land cover classification.

To assess the relationship between land cover types and evapotranspiration, we divide the study area into precipitation zones based on annual mean precipitation in the Ordos Basin. Subsequently, we refine the differences in instantaneous, monthly, and annual evapotranspiration across different vegetation types within these precipitation zones, as visualized in Fig. 10. Triangles represent extreme evaporation values (the left triangle indicates the minimum value, and the right triangle indicates the maximum value), while circles denote the mean evapotranspiration. Green, red, and blue correspond to the forest, cropland, and grassland vegetation types, respectively.

At the instantaneous scale [see Fig. 10(a)], forested land consistently exhibits higher evapotranspiration than cropland and grassland. While cropland and grassland evapotranspiration values align more closely, the discrepancy between them widens with increasing precipitation. Analyzing monthly variations in evapotranspiration over six selected months (January, March, May, July, September, and November) in Fig. 10(b)–(g), we observe that forest consistently records the highest evapotranspiration, with cropland and grassland displaying varying dynamics. In regions receiving less than 400 mm of rainfall, evapotranspiration remains relatively stable over time. In January, grassland evapotranspiration surpasses cropland, but, during the crop-growing season from March to September, cropland exhibits higher evapotranspiration. After September, grassland surpasses cropland once again. In areas with precipitation exceeding 500 mm, grassland consistently exhibits higher evapotranspiration than cropland although, during the summer crop growth season, cropland evapotranspiration temporarily surges due to rapid crop growth. Nevertheless, grassland evapotranspiration ultimately prevails. This analysis highlights distinct seasonal variations in evapotranspiration among vegetation types [43].

At the annual scale [see Fig. 10(h)], forested land boasts annual evapotranspiration exceeding 500 mm, while cropland and grassland range between 450–500 and 450–550 mm, respectively. In regions with over 600 mm of precipitation, the annual evapotranspiration gap between cropland and grassland can exceed 40 mm, widening as precipitation increases. Generally, higher precipitation levels provide more abundant atmospheric water vapor, facilitating increased evapotranspiration. It is noteworthy that when precipitation falls below 300 mm, evapotranspiration remains high, often nearing or even surpassing precipitation levels in the 500–700-mm range. This anomaly is particularly pronounced in areas with less than 300 mm of precipitation, such as the northwestern region, a significant agricultural base known as the Ningmeng Irrigation District. Here, extensive irrigation practices significantly impact evapotranspiration rates. In summary, our analysis underscores notable discrepancies in evapotranspiration across different vegetation types, with these disparities becoming more prominent with higher precipitation levels. In addition, various vegetation types exhibit distinct evapotranspiration patterns throughout the year, emphasizing the value of considering multiperiod evapotranspiration as a supplementary input feature for classification.

Comprehensively examining the relationships among NDVI, evapotranspiration, and land cover, we explore whether variations in evapotranspiration exist among different vegetation types within NDVI intervals. The results, as presented in Fig. 10(i), reveal the interval range of evapotranspiration,

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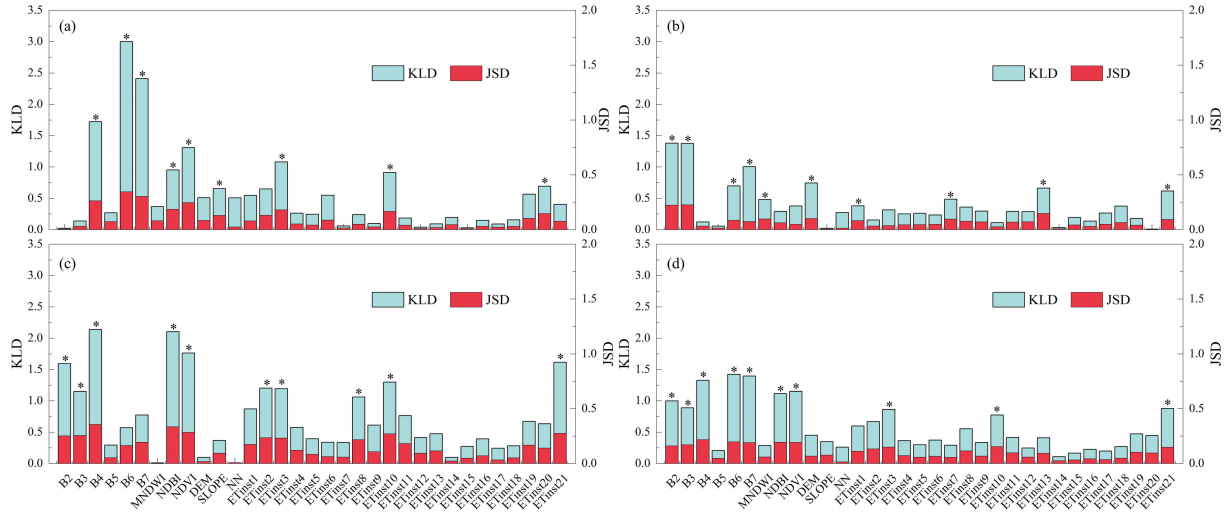


Fig. 11. Comparison of the difference of input features in different vegetation types. (a) Cropland and forest. (b) Cropland and grassland. (c) Forest and grassland. (d) Cropland, forest, and grassland. Note: * refers to input features ranked in the top ten of KLD or JSD.

representing the distance between identical color triangles along the same line. Forest exhibits the narrowest range, followed by cropland and grassland, indicating that evapotranspiration values are more concentrated in forested areas. The most significant difference in evapotranspiration distribution occurs when NDVI is below 0.5, with cropland, forest, and grassland exhibiting average evapotranspiration values distributed across the ranges [188.2 mm, 688.3 mm], [372.0 mm, 687.5 mm], and [146.3 mm, 689.8 mm], respectively. The upper boundaries (maximum values) of evapotranspiration are generally proximate, with the most pronounced difference occurring in the $0.4 < \text{NDVI} < 0.6$ range. Notably, this range poses challenges for distinguishing cropland from grassland, given their spectral similarities. The lower boundaries (minimum values) of evapotranspiration exhibit more significant variation. When $0.1 < \text{NDVI} < 0.5$, cropland and grassland differ by approximately 24.7–95.1 mm in terms of minimum evapotranspiration. Furthermore, the difference in average evapotranspiration between cropland and grassland increases from 22.9 to 49.2 mm as NDVI surpasses 0.5. This NDVI range, characterized by challenges in vegetation type classification, underscores the effectiveness of introducing evapotranspiration as a complementary feature.

In conclusion, our analysis demonstrates that evapotranspiration exhibits varying magnitudes and annual patterns across different vegetation types. This validates the utility of incorporating evapotranspiration into land cover classification as an additional and effective information source, ultimately enhancing classification accuracy. Furthermore, our findings affirm the feasibility of the proposed methodology.

2) Comparison of Input Differences in Vegetation Types: To elucidate the reasons behind the improved classification accuracy achieved by integrating evapotranspiration, we conduct a comparative analysis of input feature differences among different vegetation types using the KLD and JSD metrics. This analysis identifies key input features contributing to the enhancement of classification accuracy.

Fig. 11 illustrates the differences in the probability distributions of input features between cropland and forest, cropland and grassland, forest and grassland, and all three vegetation types combined. These probability distribution differences reflect variations in original input features across different vegetation types. Greater differences indicate more significant contributions to classification accuracy.

The results from both KLD and JSD metrics align consistently. The features exhibiting substantial differences between cropland and forest [see Fig. 11(a)] include B6, B7, B4, NDVI, ET_{inst3}, NDBI, $\text{ET}_{\text{inst10}}$, $\text{ET}_{\text{inst20}}$, SLOPE, and ET_{inst2}. Similarly, between cropland and grassland [see Fig. 11(b)], features with significant differences encompass B2, B3, B7, DEM, B6, $\text{ET}_{\text{inst13}}$, $\text{ET}_{\text{inst21}}$, ET_{inst7}, MNDWI, and ET_{inst1}. Finally, in the comparison between forest and grassland [see Fig. 11(c)], the features exhibiting notable differences include B4, NDBI, NDVI, $\text{ET}_{\text{inst21}}$, B2, $\text{ET}_{\text{inst10}}$, ET_{inst2}, ET_{inst3}, B3, and ET_{inst8}. Notably, the spectral band characteristics, such as B6, B7, B4, NDVI, NDBI, B2, B3, $\text{ET}_{\text{inst21}}$, ET_{inst3}, and $\text{ET}_{\text{inst10}}$, consistently exhibit substantial differences among the various vegetation types. This highlights that the intrinsic spectral band information effectively distinguishes between vegetation types. Additional sequential ET_{inst} as information also contributes to the differentiation and variation of vegetation types. ET_{inst} in different periods presented the variability in evapotranspiration characteristics related to vegetation growth. Consequently, the incorporation of supplementary information can enhance classification accuracy.

To further analyze the impact of various input features on vegetation type differentiation, we visualize the distribution of sample points across 3-D space for selected input features using high KLD or JSD values. These visualizations are presented in Fig. 12. In Fig. 12(a), we explore the distribution of sample points based on spectral bands B2, B3, and B5. Forested areas appear relatively independent, while cropland and grassland points exhibit substantial overlap, indicating limited input feature differences that hinder classification. Fig. 12(b) extends this analysis to include three periods

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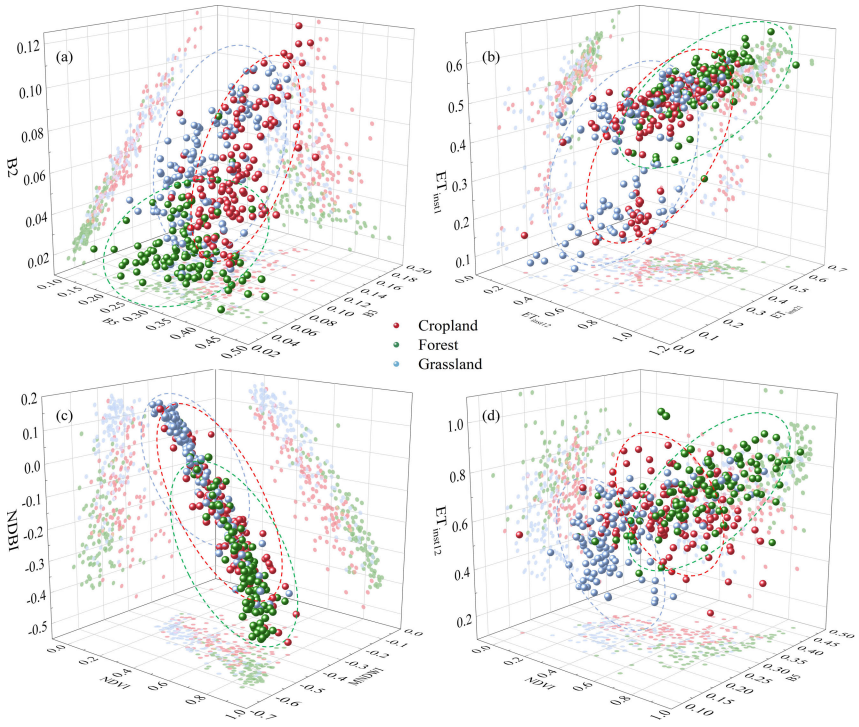


Fig. 12. Aggregation of sample distribution under different combination types of input features. (a) Spectral bands. (b) Instantaneous evapotranspiration. (c) Spectral index. (d) Combination of spectral bands, instantaneous evapotranspiration, and spectral index.

of ET_{inst}, namely, ET_{inst1}, ET_{inst12}, and ET_{inst21}, yielding a pattern similar to Fig. 12(a). In Fig. 12(c), sample points are distributed based on three spectral indices: NDVI, NDBI, and MNDWI. In this case, point groups cluster closely and largely overlap, signifying that classification solely based on spectral indices may struggle to distinguish between vegetation types effectively.

Finally, we consider spectral band B5, spectral index NDVI, and ET_{inst12} together and display the 3-D distribution of sample points in Fig. 12(d). This combination illustrates an enhanced divergence among the point groups representing cropland, forest, and grassland. In essence, by incorporating instantaneous evapotranspiration alongside traditional input features, we augment the divergence between these vegetation types, leading to improved classification accuracy.

V. Discussion

A. Consistency and Applicability Analysis of Multisource LULC Datasets in Mixed Regions of Multiple Vegetation Types

The availability of numerous land cover products worldwide offers valuable insights, yet discrepancies often arise due to differences in spatial and temporal resolutions, leading to challenges in achieving consistency and applicability. These discrepancies are particularly evident in regions encompassing diverse vegetation types [11], [78].

Notably, [68] and [79] underscore the limitations of global-scale products in accurately characterizing vegetation types, especially when large scale products are applied at finer regional scales. Similar inconsistencies have been observed in regional assessments, such as [80] evaluation of cropland products in China and [81] comparison of global products in North China. Thus, the selection of land cover products for specific study areas requires careful consideration of regional characteristics to ensure both high applicability and precision.

The challenge of scale, often referred to as the scale problem in remote sensing [83], becomes particularly pronounced when large-scale land cover products are applied to smaller spatial scales. Such applications can result in issues such as reduced spatial resolution, interannual instability, spatial discontinuity, and increased uncertainty. Land cover heterogeneity is a fundamental driver of these problems, necessitating adjustments and enhancements in classification methods to accommodate regional characteristics effectively [17], [46].

As mentioned, challenges persist in vegetation classification due to spectral similarities and mixed pixels, especially in mixed vegetation regions. These challenges are attributed to the similarities in vegetation shape and reflectance, as well as the influence of weather conditions, such as clouds and sunlight, on vegetation reflectance [84]. To address these issues and enhance classification accuracy, it is common practice to incorporate additional information as input features. However, the selection and combination of these features, particularly from a spectral perspective, have not been definitively established in the literature. While many studies have explored the impact of environmental factors on vegetation change, focusing on meteorological and topographic factors [31], few have investigated the relationship between evapotranspiration and LULC (LUCC). This study introduces evapotranspiration as a novel input feature for land cover classification, especially in mixed vegetation regions where conventional methods may fall short.

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B. Contribution of Evapotranspiration Spatial Heterogeneity to Vegetation Classification

Why actual evapotranspiration is used as an additional feature input to improve vegetation classification accuracy in this article? Can the spatial difference of actual evapotranspiration indicate the spatial heterogeneity of vegetation distribution? In previous studies, the additional information on vegetation classification is determined by analyzing the driving factors of LUCC. Generally, the driven factors can be divided into two kinds, including natural factors and human activities. The natural factors are topography, precipitation, temperature, sunlight, soil, and so on. Human activities refer to urban expansion, cropland reclamation demand, animal interference, and so on [32], [33], [85]. The factors of human activities are relatively difficult to measure accurately, which is only a macro evaluation. At present, more attention is paid to natural factors. Topography and precipitation are the main factors influencing the land cover change [26], [30]. Most of the studies focused on the analysis of the above environmental factors. However, vegetation distribution might be different under a similar environment even though, in a different environment, the vegetation might also be similar. Relatively few studies analyzed the relationship between evapotranspiration and LULC or vegetation. For example, Meng et al. [86] obtained the conclusion that evapotranspiration was the most important driven factor of LUCC in the Loess Plateau. It is also suggested that the evapotranspiration should be adjusted by reasonable allocation of land cover types, so as to allocate water resources.

The correlation between evapotranspiration and LULC is tested by statistical method. In fact, evapotranspiration is the reflection of LULC, and the spatial differentiation of evapotranspiration can characterize the spatial difference of LULC, which is proved in the study of Zhang et al. [87]. It showed that vegetation greening in the Yangtze River Basin is the main reason for the significant increase in annual evapotranspiration. Zheng et al. [88] showed that leaf area is the main reason for the impact of evapotranspiration. Scholars have also conducted a lot of research on similar research areas, such as the Loess Plateau. Ordos Basin is located in the core area of the Loess Plateau, so its geographical attributes are very similar to some extent. The study of the Loess Plateau can be used for reference to the Ordos Basin. Zhang et al. [87] reported that vegetation restoration in the Loess Plateau increased evapotranspiration. Yang et al. [41] quantitate that the effects of vegetation greening on soil evaporation, vegetation transpiration, and evapotranspiration on the Loess Plateau were −16.6, 20.7, and 4.1 mm/year, respectively. Therefore, a great number of papers have shown that vegetation changes will cause evapotranspiration change [15], [89], [90].

Evaporation is the process of water vapor escaping from the surface to the atmosphere, and it is called transpiration in vegetation. It refers to the process by which plants absorb water from the soil and transport it to stems and leaves, and most of the water vapor escapes from the leaves and stems into the air. Air temperature, land temperature, and net radiation are the thermal factors that affect the actual evapotranspiration. Soil water moisture is the water condition, and wind speed is the dynamic factor. The evapotranspiration of different vegetation or land cover types all show significant spatial variability, even at the same meteorological conditions [91]. The evapotranspiration of forest, cropland, and grassland accounted for the largest proportion of total evapotranspiration [74]. Specific to different vegetation types, the largest one is forest, followed by cropland and grassland [39], [43]. For example, Dong et al. [40] simulated the actual evapotranspiration in China based on the SPAC model; the result showed evapotranspiration in forests was the largest (451–512 mm/year), followed by croplands (387 mm/year), grasslands (228 mm/year), and barren (151 mm/year). According to research by Zhang et al. [38], the evapotranspiration of grassland is greater than that of cropland. Yang et al. [41] also pointed out that the change of grassland or cropland to forest and the change of cropland to grassland would increase evapotranspiration.

In Fig. 10, the differences in instantaneous and annual evapotranspiration were compared in similar meteorological conditions. The mean values of ET_{inst} in forest, cropland, and grassland are 0.56, 0.42, and 0.41 mm. On the annual scale, the annual evapotranspiration in forests is more than 500 mm, which, in cropland and grassland, ranges from 450 to 500 and 450 to 550 mm. In general, the increase in vegetation will reduce the SR, receive more net radiation flux, increase the aerodynamic resistance, and increase the air humidity and evapotranspiration [88]. On the contrary, in areas with low vegetation coverage, water restriction increases, and the absorbed energy is mainly used to heat the air, leading to temperature rise. Heat flux accounts for a large proportion of net radiation flux, and evaporation is relatively small [92]. Among the three vegetation types, cropland, forest, and grassland, the forest has the deepest root system, the best water conservation ability, high soil moisture retention, the largest canopy, the strongest transpiration, the highest water consumption, and the highest evapotranspiration. The root system of cropland and grassland is not as deep as that of forest, the plant height is relatively small, the aerodynamic resistance is small, and the evapotranspiration is relatively small and close. In Fig. 3, ET_{inst} of the three vegetation types increased first and then decreased in 2020. The annual distribution process of evapotranspiration presents a single peak pattern, reaching its peak in July. There is a significant difference between forest land and the other two vegetation types, while there is a small difference between cropland and grassland. During the crop growth period, evapotranspiration of cropland is greater than that in the grassland, while, in other periods, it is basically flat or slightly less than the evaporation of grassland [93]. This is because the evapotranspiration of grassland is slightly larger than that of cropland in autumn and winter, while, in summer and late spring, the evapotranspiration of cropland is larger than that of grassland, which is related to the regional agricultural planting system [94]. The crops in the Ordos Basin are mainly corn and wheat, and most of the crop growth period is between April and October. Late spring and summer are the peak seasons of crop growth due to sufficient hydrothermal conditions, leading to rapid growth of vegetation leaves and larger evapotranspiration. In late autumn and winter,

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the decrease in temperature and sunshine time resulted in a decrease in net radiation flux in autumn and winter. At the same time, the contribution of crops to evapotranspiration decreased when they entered the mature and aging stages, resulting in a decrease in evapotranspiration [71].

Previously, NDVI was considered in vegetation classification, but it was found that the NDVI will also change in areas with unchanged land cover. NDVI may be better at distinguishing vegetation from nonvegetation, and there are also problems in subdividing different vegetation types. We discussed this problem in Fig. 9. Almost all vegetation types are distributed on each NDVI interval, which plays a limited role in specific vegetation classification. The NDVIs of cropland, forest, and grassland are about 0.5–0.8, 0.4–1.0, and 0.2–0.7. There is a certain overlap in the distribution range of NDVI. It indicates that when NDVI is a certain value, there may be multiple vegetation types on it; only the probability is different. This will cause some interference in vegetation classification. However, when the evapotranspiration factor is considered together, the results are shown in Fig. 10(i). As NDVI increases, the difference in average evapotranspiration between cropland and grassland gradually increases from 22 to 50 mm. To some extent, it can compensate for the differences in vegetation characteristics in NDVI easily confused areas ($0.4 < \text{NDVI} < 0.9$). The results show that the evapotranspiration of cropland, forest, and grassland varies greatly in different climatic regions in magnitude and annual evolution characteristics. The evapotranspiration will show greater differences in different NDVI and land cover types, which proves that using evapotranspiration in land cover classification is effective information and can improve classification accuracy [86]. At the same time, it also demonstrates the feasibility of using evapotranspiration differences to improve the classification accuracy of vegetation types [91].

C. Regional Applicability and Limitations of Land Cover Mapping Based on Evapotranspiration

This research highlights the potential of evapotranspiration as a valuable input feature for land cover classification, particularly in mixed vegetation regions. By incorporating this spatially heterogeneous information, we can improve the accuracy and applicability of land cover mapping, addressing long-standing challenges associated with vegetation classification. In previous classification applications, the traditional input features may have produced satisfactory results. However, there is a challenge due to the similarity of spectral characteristics in areas with mixed vegetation [30]. To improve classification accuracy in such areas, the method of considering the spatial differentiation of actual evapotranspiration can be effectively applied. The vegetation area still accounts for the largest proportion, and the distribution is often irregular and mixed in nature [3], [45], [95]. The spatial difference of vegetation evapotranspiration was taken into account to characterize the heterogeneity of vegetation spatial distribution in this article. The evapotranspiration is inducted from the real environment [89]. Vegetation evapotranspiration attributes innovatively to improve land cover classification accuracy, especially in vegetation classification. The research method constructed in this article has important promotion prospects and universal significance in the mixed vegetation area. Ordos Basin is a typical climatic transition and vegetation mixing region. The method introduced in this study is not limited to the Ordos Basin but is adaptable to a wide range of vegetation distribution scenarios. In the future, this new method can be extended to a larger scale for further verification and comparison. Furthermore, vegetation types can be further refined. Plants of different subjects and genera exhibit varying properties in evapotranspiration based on their unique physiological characteristics. Therefore, this method can also be applied in future research to identify vegetation types more accurately.

The limitations of these studies are concluded as five points as follows. First, the remote sensing images of Landsat at a 30-m resolution are applied here. At present, the spatial accuracy of remote sensing products is becoming more and more precise, but it is not the higher the resolution, the better the application. On the one hand, higher resolution means higher data volume and computation. On the other hand, the resolution of remote sensing data should match the actual application scale of hydrological, ecological, and other related models [19]. Second, the accuracy of evapotranspiration calculation affects the accuracy of classification results. The meteorological data in this article are in a station scale, while LST, NDVI, and albedo are scaled uniformly data with different spatial resolutions initially. Input data with the same spatial resolution and finer resolution can be applied in the future. Moreover, SEBAL is a single-source evapotranspiration model; the double-source evaporation evapotranspiration model can be applied for higher precision in the future [96]. Third, six land cover types were set to classify in this article. The classification system can be further refined in the future to enhance the recognition of land cover. Fourth, increase the sample size and use cross-validation methods to avoid overfitting issues. Finally, the ETLULC framework has a significant variance and improvement compared with previous input schemes. Then, ETLULC was compared with six existing products based on the suggestions to avoid the interference of sample selection to comparison in different dimensional assessments. That is, the consistency and variance are analyzed and compared by consistency test or difference degree test. Ordos Basin is a typical climatical transition and vegetation mixing region. In the future, this new method can be extended to a larger scale for further verification and comparison.

In conclusion, this research highlights the potential of evapotranspiration as a valuable input feature for land cover classification, particularly in mixed vegetation regions. By incorporating this spatially heterogeneous information, we can improve the accuracy and applicability of land cover mapping, addressing long-standing challenges associated with vegetation classification.

VI. Conclusion

In this study, we have introduced a novel method aimed at enhancing land cover type identification within mixed regions characterized by diverse vegetation types, notably in forest–cropland–grassland/shrub mixed areas. The objective

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of this method is to address the challenges associated with distinguishing between various vegetation types, referred to as ETLULC. ETLULC exhibits an OA of 93.1%, with PA, UA, and $F1$-score ($F1$) for all land cover types surpassing 85% and 90%, respectively. Furthermore, we observed that integrating evapotranspiration data improved the OA by 7.4%–14.2%, while the accuracy of classifying grassland, forest, and cropland improved by 3.0%–20.4%, −0.7% to 15.7%, and 5.4%–15.3%, respectively.

Our findings indicate that ETLULC exhibits varying degrees of consistency with existing land cover products. We also detected omissions of cropland, forest, and grassland ranging from 17% to 45%, 28% to 68%, and 11% to 44%, respectively, when existing products were applied to the Ordos Basin. These errors resulted in an overestimation of cropland, an underestimation of forest, and an overestimation of grassland. Finally, we explored the differences in evapotranspiration among different vegetation types and parameters (e.g., NDVI) under identical meteorological conditions. Forests exhibited the highest evapotranspiration levels, characterized by a concentrated distribution range. Cropland and grassland had slightly lower evapotranspiration rates, with variations over time. Notably, with increasing precipitation, the annual evapotranspiration gap between cropland and grassland averaged over 40 mm. In addition, the introduction of evapotranspiration as an input feature increased the differences in traditional input features among vegetation types. These conclusions underscore the effectiveness of incorporating evapotranspiration data into land cover classification for improved accuracy.

From a computational and principled standpoint, the method introduced in this study is not limited to the Ordos Basin but is adaptable to a wide range of vegetation distribution scenarios. The land cover mapping framework developed in this study holds significant promise for application in various mixed vegetation regions, with potential for further optimization and validation. Consequently, it carries important implications and broad applicability within the field of remote sensing.

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